Re-engineering: JWST Subsystem Modification

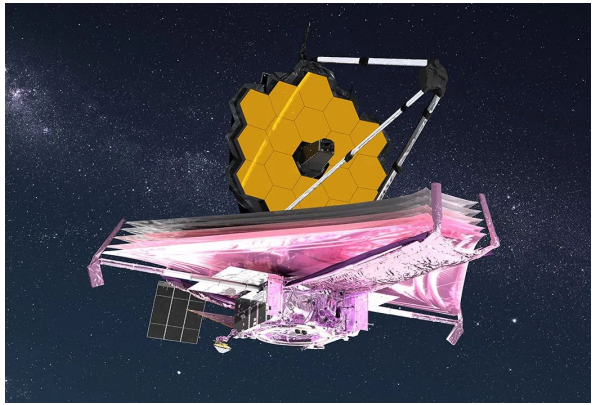
By Melanie Ye and Francesca Mangano
January 24th, 2026

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Problem statement

The James Webb Space Telescope (JWST) takes high-resolution infrared photos of the universe's earliest galaxies, stars, and exoplanets.



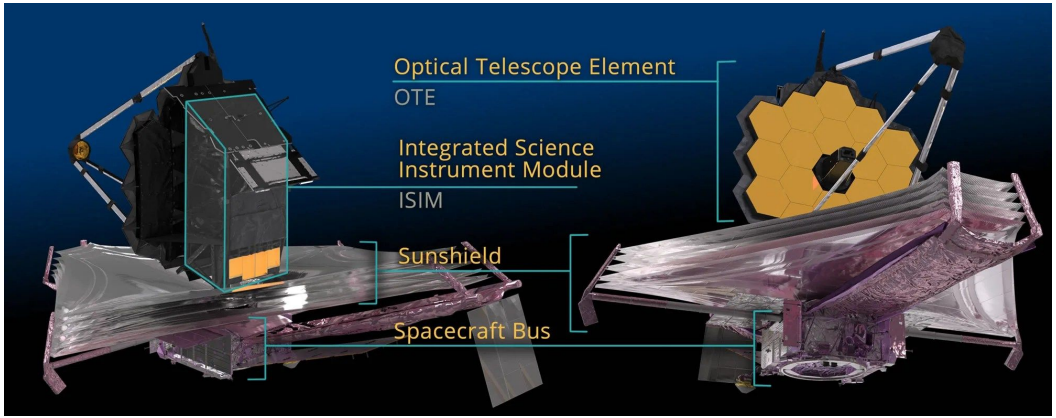


However, it is limited by its finite supply of onboard propellant…

Re-engineering the bus system

Problem: If the JWST wants to have a successful mission to orbit Pluto, it must have a propulsion system invested in its longevity





This design is centered around the bus and its propulsion system

In scope	Out of scope
• Spacecraft bus • Delta V orbit correction propulsion • Fine tuning propulsion • Power generation • Heating/cooling	• Getting the modified JWST in Pluto's orbit • All other spacecraft elements

Functions and objectives

Primary function:

The design must transform the JWST’s current bus into an efficient, sustainable system designed for travel orbiting pluto

4 main objectives:

1. Promote long-term sustainability
2. Maximize power efficiency
3. Achieve thermal stability
4. Remain reliable for a 10+ year mission

Regulatory constraints

- Adhere to… [source]
  - “Moon agreement” for peacekeeping
  - Convention on International Liability for Damage Caused by Space Objects
  - Convention on Registration of Objects Launched into Outer Space
- Safety
- Approval from the INSRB
- Nuclear power approval

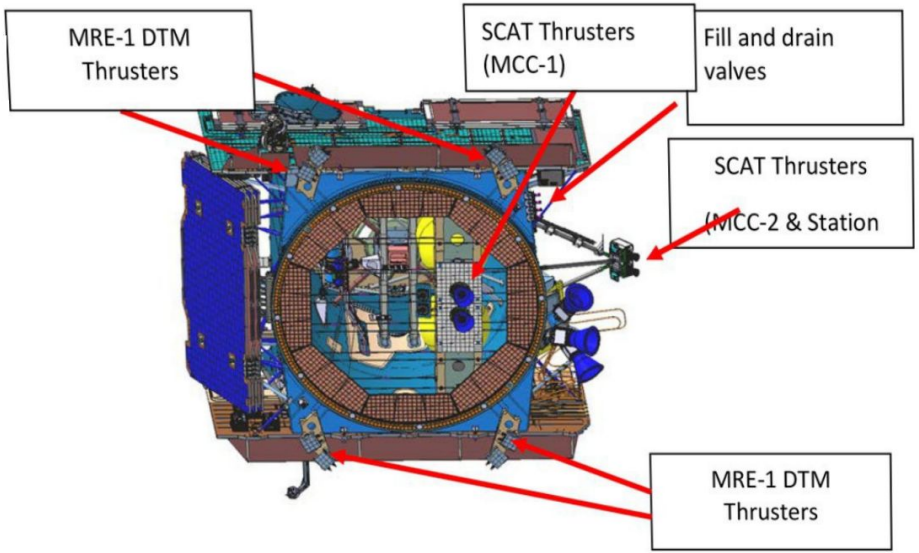
This design must follow all international space laws and conventions regarding both space travel and nuclear energy

Design constraints

Thrusting capacity	11.5 m/s of thrust	• Thrusting capacity will be for orbit correcting Delta V maneuvers • Thrust capacity was modele based on the ion propulsion system that orbits dwarf planet Ceres [source]
Nuclear energy capacity	At least 9 kW	• Chosen ion thruster requires 7.4 kW to operate [source] • Upper bounded to account for other bus energy requirements

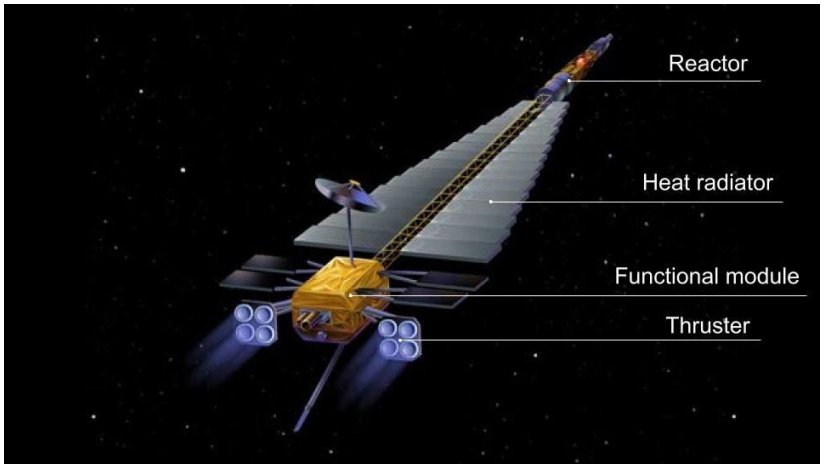
JWST's current propulsion system

Good enough for L2, but highly inefficient for deep space travel

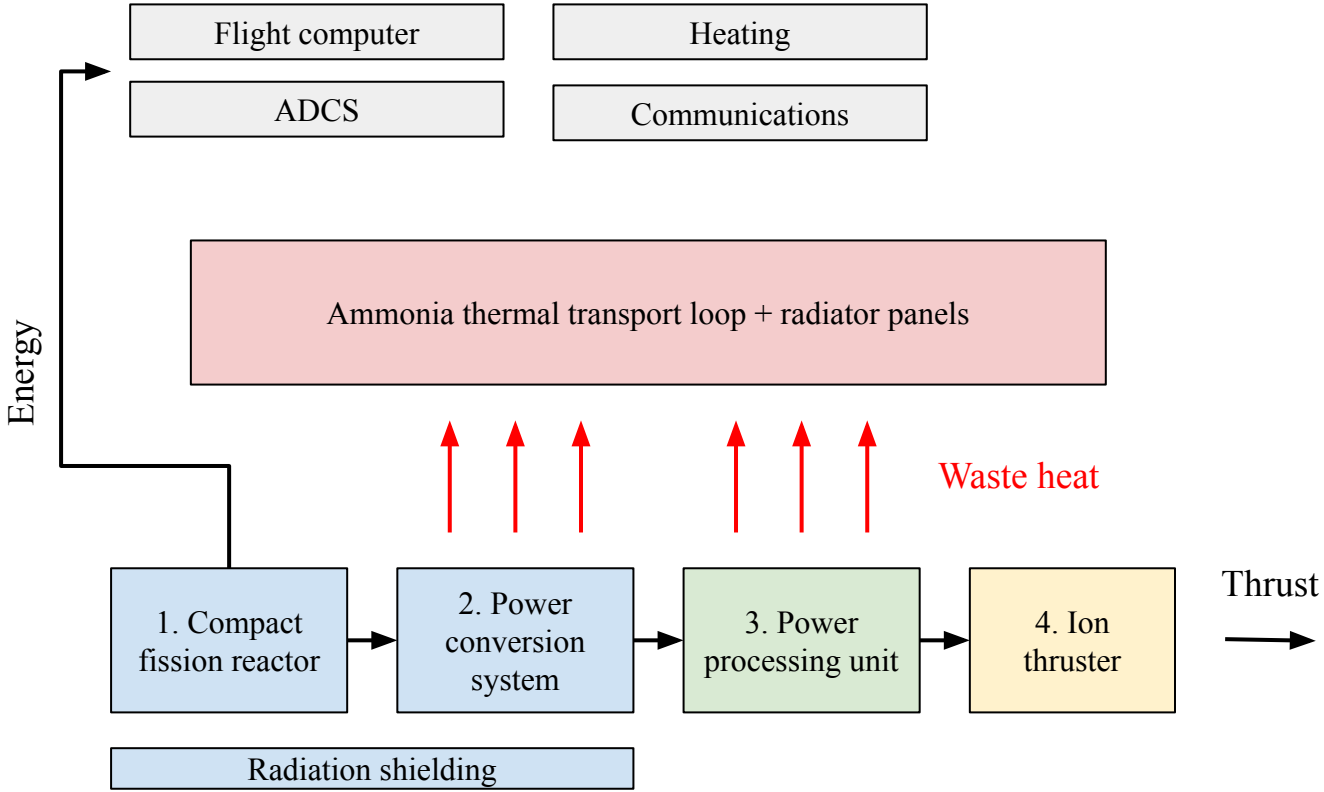


Design approach

What technology can scale the JWST's propulsion from L2 to Pluto's orbit?



Proposed design: The NovaBus



1. Compact fission reactor



Figure 6. KRUSTY. Source [x]

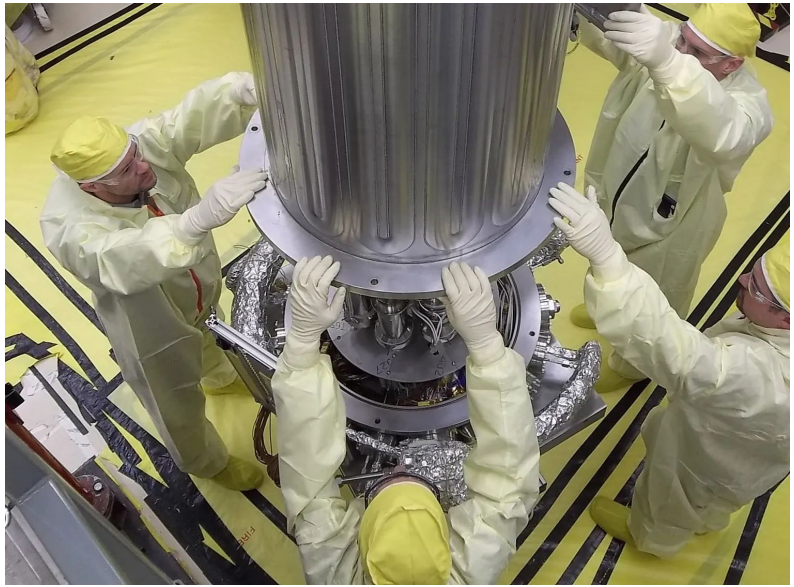
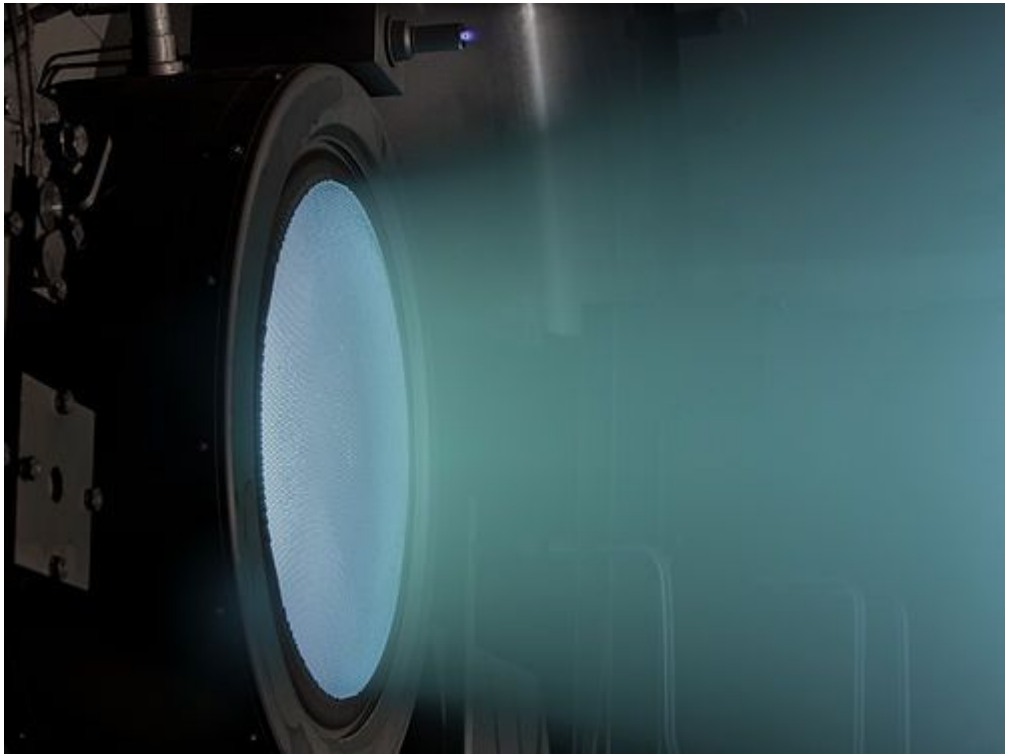


Figure 7. KRUSTY testing. Source [x]

2. Power conversion system and 3. PPU

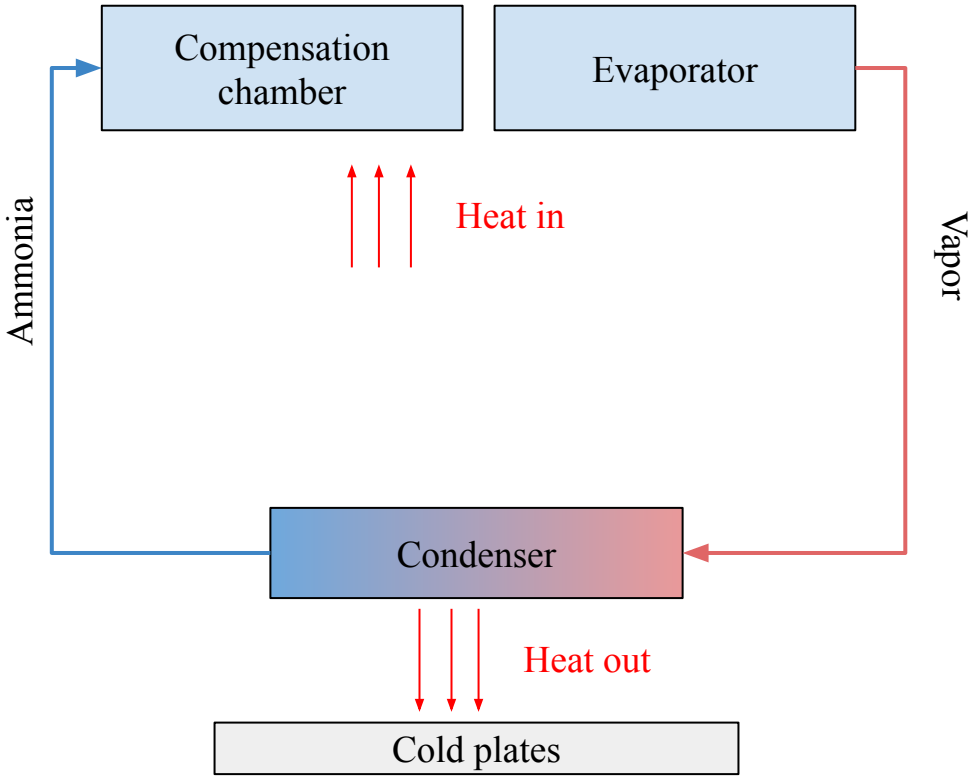
2. Power conversion system	1.1 Power conversion unit: Converts thermal power into shaft power using sodium heat pipes 1.2 Power conversion: System converts AC to DC 1.3 Power distribution: Regulated DC power operates the bus load whereas high voltage DC uses the PPU (3) to operate the ion thrusters	High voltage power will be for the thrusters, whereas low voltage power will be for the flight computer, ADCS, comms, and heating
3. Power processing unit (PPU)	Converts bus power into controlled currents, throttles power	PPU creates high-current pulses of power, which allow the ion thruster to operate

4. Ion thruster



Requires <7.4 kW to operate under 4220 ISP at 70% efficiency

Ammonia radiator system



NovaBus: Weighted objective analysis

		Existing design		Proposed design	
Objective	**Weight**	**Score (/10)**	**Weighted**	**Score (/10)**	**Weighted**
1) Promote long term sustainability	0.25	3	0.75	9	2.25
Justification		Propellant limits lifespan to only 5-10 years, specific impulse is 200-285 s For material sustainability, refer to Appendix E.		NEP has a ~2000-8000 s specific impulse For material sustainability, refer to Appendix E.	
2) Maximize power efficiency	0.25	6	1.5	9	2.25
Justification		Solar panels generate 2,000 W of solar array power		Compact fission reactor that generates 40 kWe	

NovaBus: Weighted objective analysis

		Existing design		Proposed design	
	Weight	**Score (/10)**	**Weighted**	**Score (/10)**	**Weighted**
3) Achieve thermal stability	0.25	7	1.75	9	2.25
Justification		JWST's sunshield protects it from the sun's radiation at L2		Pluto's climate omits the need for sun radiation protection, ammonia thermal radiator system keep nuclear reactor from causing heat gradients on the proposed design	
4) Remain reliable for a 20+ year mission	0.25	1	0.25	9	2.25
Justification		Will only last at maximum, for 10 years		Nuclear fission reactor, which can be used at 25% to extend lifetime from 10 years to an upper bound of 40 years	
Total	1	-	4.25	-	9

Environmental considerations

Material	Environmental Impact of Production	Environmental Impact of Usage
Xenon	• High energy consumption leads to greenhouse gas emissions • Requires air separation, and the process produces pollutants	• Inert gas; does not contribute to greenhouse gases and ozone depletion
Hydrazine Fuel	• Negative health effects when exposed to humans and animals • Accidental discharge into air, soil, and water may occur during any production stage • Minimal impact because hydrazine degrades quickly in these environments	• Produces toxic emissions of ammonia, hydrogen, and nitrogen gases. • Produces greenhouse gases that trap heat in an atmosphere.
Dinitrogen Tetroxide	• Discharge into water leads to excess algae growth. • Pollution of nitrogen oxides are toxic to humans and damaging to plants	• Leakage into soil decreases soil pH • Causes changes in Ozone • Forcing agent that changes energy balance of surroundings, can contribute to climate change

Social considerations

Stakeholder	Description/Interest
Government/regulators	• Ensure compliance with international laws and safety standards
Telescope operations team (on Earth)	• Responsible for telescope operations • Use telescope communications system to operate thrusters
Space agencies	• Funds missions and projects
Material suppliers and manufacturers	• Responsible for testing and manufacturing of different materials
Space research groups	• Will be executing the testing and creation of the NovaBus

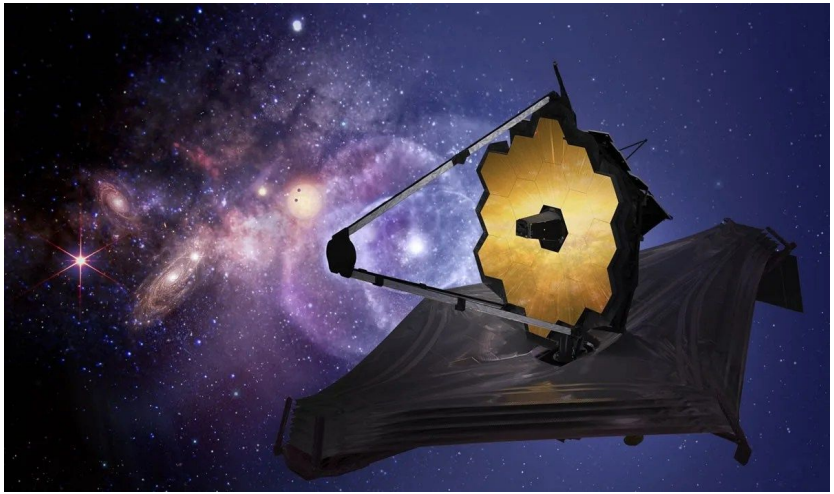
Economic feasibility

Propellant	Unit Cost	Unit Cost Adjusted for Inflation	Amount Used	Total Cost
Xenon	$1500/kg in 2001	$2,728.33/kg	155 kg	($2,728.33/kg)* (155 kg) = $422891.15
Hydrazine fuel	$143.53/lb or $65.10/kg in 2019 [x]	$81.98/kg	168 kg	($81.98/kg)*(168 kg) + ($125.23/kg)* (301 kg) = $51466.87
Dinitrogen tetroxide oxidizer	$196.89/lb or $89.31/kg in 2012 [x]	$125.23/kg	301 kg	

Conclusion

Using Tsiolkovsky's equation, it is estimated that the NovaBus will require 155 kg of Xenon fuel for its Pluto mission

Compared to the JWST's combined total of 301 kg to go to L2



Re-engineering:
DSN For Modified JWST

By Melanie Ye and Francesca Mangano
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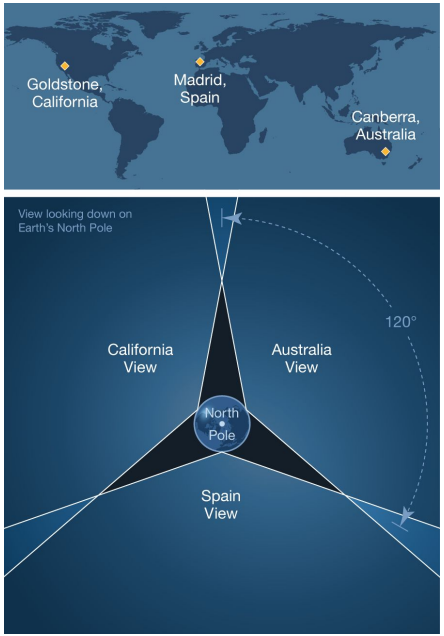
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Problem statement

Deep space networks (DSN) rely heavily on radio waves

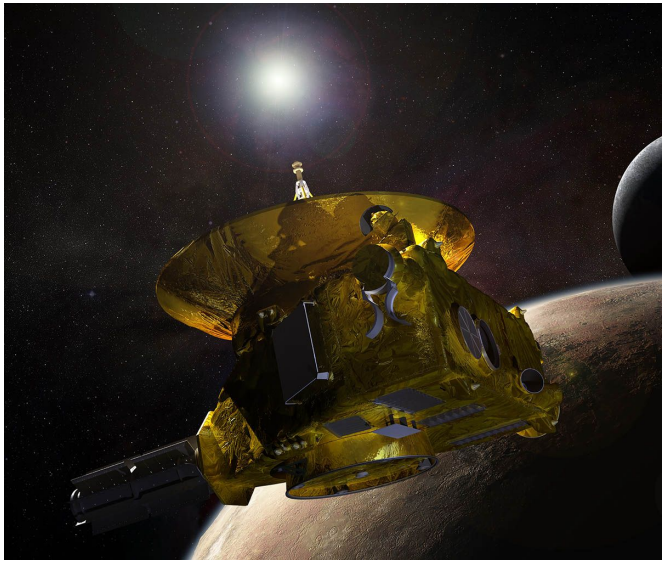
- Slow speeds
- Low transmission frequency

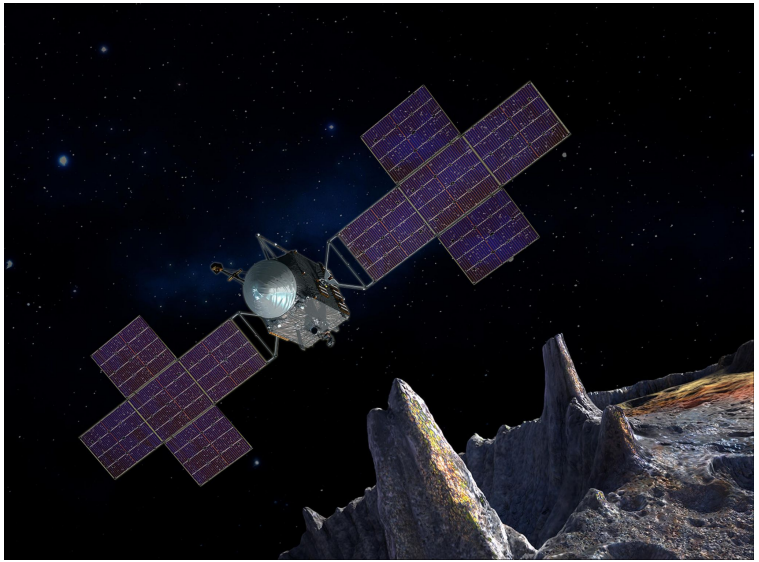




Re-engineering the DSN for the NovaBus modified telescope

The current DSN makes 1-way communication to Pluto take upwards of 5 hours





Functions and objectives

This design has 1 primary function:

Modifying the DSM network to contact the modified JWST/NovaBus telescope in as little time as possible

And 4 objectives:

1. Promote long term sustainability
2. Large data output
3. Operate with minimal interference
4. Support high wavelength frequencies

Design scope

This design must only

- Add on to existing DSM architecture
- Implement as many existing design principles as possible





Design constraints

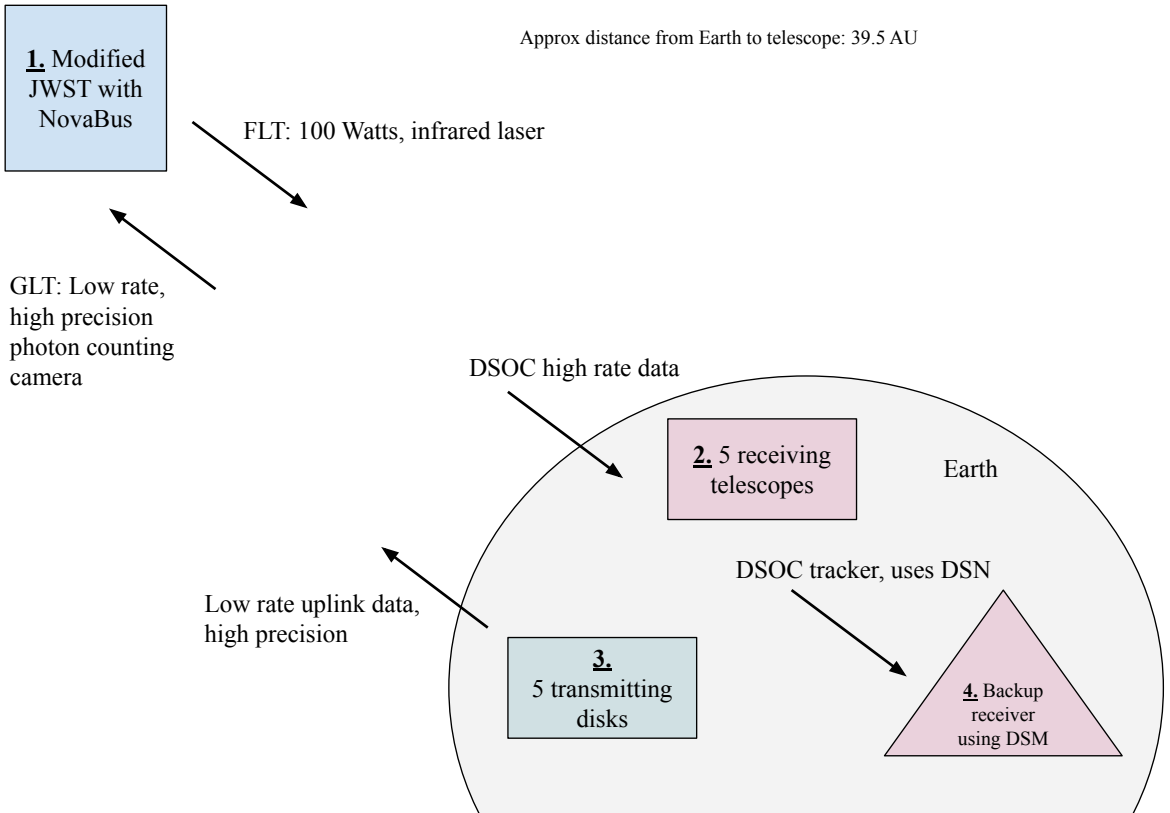
Regulations	• Follow international treaties (ISED) • Not interfere with other space communications systems	• Proposed design must follow all international space laws and conventions
Interference	• <5 signal-to-noise ratio (SNR)	• Benchmarked from the New Horizon's SNR
Transmission speed	• <5 hours one direction • Faster than 10-50 kbps	• New Horizons' speed was used as a benchmark (5 hours, 8 mins for a 3.4 billion mile distance) • Lucy mission reached an average of 10-50kbps

Design approach

Inspired by the infrastructure for the Psyche mission, the proposed design implements infrared Deep Space Optical Communications (DSOC)



Proposed solution



1. Telescope modifications

1. Modified JWST with NovaBus	1.1 Flight laser transceiver system (FLT): 1550 nm infrared laser with 100 watt transmission capacity, short pulse 1.2 Ground laser transmitter (GLT): Accurately pointed beam for photon counting 1.3 Vibration prevention system: FLT is mounted on actuators and struts to stabilize vibrations	Compared to the New Horizon's 12W transmission system, which sends 1 kbps to Earth. This is highly limited due to its generator, which has decayed to allowing only 12W of transmission vs the original 24W The NovaBus' nuclear power generation has more power capacity than that of the New Horizon's craft, causing 100W to be used for the optics system.

Receiving telescopes and transmission disks

2. 5 receiving telescopes	High-rate DSOC receiving telescopes in 5 locations within 300 mile radius.	Existing DSM scalability allows an increased amount of telescopes to correlate to higher reception rate
3. 5 transmitting disks	High precision infrared receivers, increased amount increases capacity	Benchmarked with the receiver from the Psyche mission

Weighted objectives decision matrix

		Existing design		Proposed design	
Objective	**Weight**	**Score (/10)**	**Weighted**	**Score (/10)**	**Weighted**
1) Promote long term sustainability	0.25	7	2.75	8.5	2.125
Justification		NASA has plans to upgrade the DSN infrastructure by replacing its six antennas with new ones, thus extending its life. However, this project is currently 5 years behind schedule		The NovaBus' nuclear power generator will sustainably supply power to the communications systems. A failure in power generation is what caused the New Horizon's system to only work at half its intended efficiency	
2) Large data output	0.25	3	0.75	8	2
Justification		New Horizon, a satellite near Pluto, has a download rate of 1-4 kbps		Having 100W of power to operate the FLT significantly increases the output capabilities, which is where current deep space technology is limited	

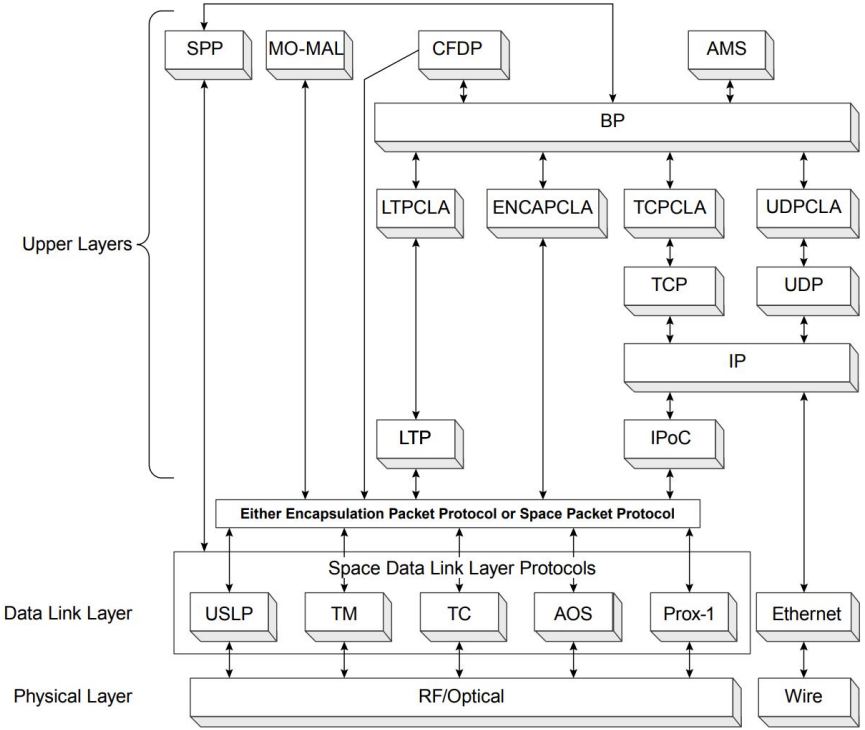
Weighted objective decision matrix

3) Operate with minimal interference	0.25	4	2	7	1.75
		Radio frequency is incredibly prone to interference (clouds, radiation) causing it to be inefficient at deep space communications		Infrared-based communication systems are still susceptible to interference and must work around it	
4) Support high wavelength frequencies	0.25	3	0.75	9	2.25
		Deep space technology, such as the New Horizons telescope transmits 1 kbps on average		Having a nuclear power source power a 100W generator is an incredible advantage that allows DSOC infrared lasers to have a wavelength of 1.55 microns	

Social implications

Stakeholder	Description/Interest
Government/regulators	• Ensure compliance with international laws and safety standards
Telescope operations team (on Earth)	• Responsible for telescope operations
Space agencies	• Funds missions and projects
Material suppliers and manufacturers	• Responsible for testing and manufacturing of different materials
Deep Space Network operators	• Engineers maintain the system 24/7, sending messages as needed.

Space communications protocols



Space communications protocols

Layer	Protocols	
Physical Layer	**Radio Frequency and Modulation Systems** • Omnibus for space links between ground stations and spacecraft. • Ex. Between spacecraft and ground stations. **Proximity-1 Space Link Protocol** • Recommendations for proximity space links. • Ex. Landers, rovers, orbiters, relays.	
Data Link Layer	**Data Link Protocol Sublayer:** Techniques for employing data units called Transfer Frames to transmit data units supplied by the higher layer via a point-to-point space link.	• **Telemetry (TM) Space Data Link Protocol** • **Telecommand (TC) Space Data Link Protocol** • **Advanced Orbiting Systems (AOS) Space Data Link Protocol** • **Proximity-1 Space Link Protocol—Data Link Layer** • **Unified Space Link Protocol (USLP)**

Space communications protocols

Synchronization and Channel Coding Sublayer:	• **TM Synchronization and Channel Coding** ○ Used with the TM or AOS Space Data Link Protocol or USLP. ○ Adds error correction and frame synchronization to TM/AOS/USLP frames for noisy downlinks. • **TC Synchronization and Channel Coding** ○ Used with the TC Space Data Link Protocol or USLP. ○ Provides BCH error-correcting coding and synchronization for TC/USLP command uplinks, with optional validation. • **Proximity-1 Space Link Protocol—Coding and Synchronization Layer** ○ Used with the Proximity-1 Space Link Protocol or USL. ○ Implements convolutional/LDPC coding and CRC for short-range bidirectional links. • **Flexible Advanced Coding and Modulation Scheme for High Rate Telemetry Applications (SCCC)** ○ For missions requiring very high data rates ○ Ex. Earth observation. • **CCSDS Space Link Protocols over ETSI DVB-S2 Standard (DVB-S2)** ○ Reduces cost and development time for high-speed downlinks.

Space communications protocols

Network Layer	• **CCSDS-recognized Internet Protocol datagrams** • **The Delay Tolerant Network architecture's Bundle Protocol** ○ Implements DTN store-and-forward routing for intermittent, long-delay space links. ○ Core protocol for deep-space, lunar, and disrupted environments.
Transport Layer	• **Space Communications Protocol Standards (SCPS) Transport Protocol** ○ Extensions to TCP/UDP for space environments. ○ End-to-end reliable transport. • **Transmission Control Protocol (TCP)/ User Datagram Protocol (UDP)** ○ Traditional transport protocols ○ Used over IP over CCSDS links • **Licklider Transmission Protocol** ○ Provides optional reliability for DTN links ○ Underlying transport for Bundle Protocol

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Space communications protocols

Application Layer	• Asynchronous Messaging Service (AMS) • CFDP • Lossless Data Compression • Image Data Compression • Multispectral & Hyperspectral Image Compression • Space Packet Protocol • Message Abstraction Layer (MAL) Space Packet Transport Binding and Binary • Encoding

Conclusion

Transmission from Psyche: $164,800 * 100/4 * 5 = $206 million

Transmission from OCTL: $205,835,200 * 5 = $1029.2 million

Transmission	Ratio of transmission power
Transmission from Psyche: 4 watt	0.0008
Transmission from OCTL: 5 kWatt	0.9992

The cost for the new proposed design is estimated to be $1.235 billion.

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